ARSHIA AKHAVAN

Master's Student in Computer Science

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EDUCATION

M.S. in Computer Science

San Diego State University (2025 – 2027) Overall GPA: 4.0/4.0

B.Sc. in Computer Engineering

Sharif University of Technology (2018 – 2024) Overall GPA: 18.96/20

Thesis: *ARK, a censorship-resistant Layer-4 VPN protocol* (advisor: Dr. Mahdi Jafari Siavoshani)

RESEARCH INTERESTS

- Programming languages and verification for concurrent and distributed systems: making deadlocks, data races, memory-ordering and consistency bugs unrepresentable
- Type systems and verified program rewriting (e-graphs, equality saturation)
- Safe systems programming in Rust: lock-free concurrency and safe memory reclamation

PUBLICATIONS

Conference Proceedings

- **A. Akhavan**, A. Hoseinpour, A. Heydarnoori, H. Bagheri, and M. Keshani, "LinkAnchor: An autonomous LLM-based agent for issue-to-commit link recovery," in *Proceedings of the ACM on Software Engineering*, FSE '26, vol. 3, ACM, 2026, pp. 4184–4207. DOI: 10.1145/3808191

Preprints

- **A. Akhavan**, E. Bartoli, A. Algharbi, A. Hoseinpur, I. Leite, and B. Donyanavard, *Yggdrasil: A layer-first 3D scene graph for real-time querying*, 2026. arXiv: 2609.38640 [cs.RO].
- A. Khosravani, A. Hoseinpour, **A. Akhavan**, M. Keshani, and A. Heydarnoori, *LIA: Supervised fine-tuning of large language models for automatic issue assignment*, 2026. arXiv: 2601.01780 [cs.SE].
- K. Kiashemshaki, A. Khosravani, A. Hosseinpour, and **A. Akhavan**, *Automated bug triaging using instruction-tuned large language models*, 2025. arXiv: 2508.21156 [cs.SE].

RESEARCH EXPERIENCE

Graduate Research Assistant

San Diego State University (advisor: Prof. Bryan Donyanavard)

Sep 2025 – Present San Diego, CA

Building **Yggdrasil**, a 3D scene graph for robots, written in **Rust** and designed for both construction and querying. Its layers form a DAG rather than a stack, and it answers pattern-matching, nearest-neighbor, field-of-view and traversal queries in place, using per-layer and per-node spatial indexes it keeps current as the graph changes. Every measured query takes 2–127 µs, up to **121× faster** than a published C++ baseline. Yggdrasil is integrated into three published robotics pipelines and it removes up to **99%** of the time they spend on their scene graph operations.

Currently extending Yggdrasil to multi-agent settings where each agent contributes its own view of the world, and a global server resolves the **logical and semantic conflicts** between them. Fully funded by NSF award CNS-2348349.

Research Collaborator — AEBench: Benchmarking AI Agents on Artifact Evaluation

University of Illinois Urbana-Champaign (with Bogdan Alexandru Stoica)

Jun 2026 – Present Remote

AEBench measures how effective AI agents are in reproducing the results of published systems artifacts, and proposes a systematic framework for evaluation and execution of AI agents on artifact evaluation. The goal is to provide Artifact Evaluation Committees with a tool to fully automate the evaluation process via use of AI agents.

Graduate Researcher — LLM Agents for Software Engineering

Bowling Green State University (advisor: Prof. Abbas Heydarnoori)

Jan 2025 – Sep 2025 | Bowling Green, OH

Built **LinkAnchor**, an autonomous agent that recovers links between issues and the commits that fix them (FSE '26, first author); **LIA**, supervised fine-tuning of LLMs for automatic issue assignment; and bug triaging with instruction-tuned LLMs.

HONORS & AWARDS

Presidential Graduate Research Fellowship (PGRF)
University merit fellowship

Sep 2025 – Dec 2027 | SDSU, CA

Awarded by SDSU for excellence in research and academic achievement, in addition to my NSF-funded research assistantship.

Konkour
118th place

Aug 2018

Ranked 118th in the National University Entrance Exam on Mathematics (Konkour) among more than 144,000 students nationwide. (Top 0.08%)

TEACHING EXPERIENCE

- **Operating Systems** (Dr. Mehdi Kharrazi) — Head TA Spring 2023; TA Spring 2022
- **Multi-Core Computing** — Head TA Spring 2023 and Fall 2023 (Dr. Hajar Falahati); TA Spring 2022 (Dr. Hajar Falahati, Dr. Hamid Sarbazi)
- **Fundamentals of Programming** — Head TA Fall 2021 and Fall 2022 (Dr. Mohammad Amin Fazli); TA Fall 2020 (Dr. Fakoori) and Fall 2019
- **Advanced Programming** (CE department) — Head TA Spring 2021; TA Spring 2020
- **Computer Networks** (Dr. Mahdi Jafari Siavoshani) — TA Fall 2022 and Spring 2023
- **Compiler Design** (Mr. Mohammad Reza Bahrami) — TA
- **Automata Theory** (Dr. Ali Movaghar) — TA Spring 2023
- **Computer Structure and Language** (Dr. Laleh Arshadi) — TA Spring 2021
- **System Analysis and Design** (Mr. Alireza Agha Mohamadi) — Head TA of the course project
- Also TA for **Numerical Computation** (Spring 2023), **Database Design** (Fall 2022) and **Big Data** (Spring 2023)
- Course design: **Zero to Hero Linux**, Quera (link)

SERVICE & OUTREACH

Artifact Evaluation Committee Member

USENIX Annual Technical Conference (ATC '26)

2026

Founder & President

Code o She'r (1st edition)

Apr 2024

Code o She'r is a Spaghetti Code Challenge that focuses on undefined behavior and exploits architectural behaviors of computers and compilers to help contestants become more familiar with the field of computer systems.
As the Founder and President of Code o She'r, I took on the role of leadership over a team of 70+ members and oversaw various flows from design, marketing, and executive duties to scientific and technical tasks.

Chief of Infrastructure and System Team

Sharif AI Challenge 2021

Dec 2020 – May 2021

The Sharif AI Challenge is an annual AI competition organized by the SSC, where teams design AI agents to compete in a game-based problem. The event features both online and on-site phases and attracts hundreds of student participants from across the country. I led the Infrastructure team to design and develop a **scalable** competition platform that could **dynamically** adjust resources based on demand while ensuring **fairness** between competitors. GitHub

Chief Technical Staff

ICPC Asia-West Continent Final Contest

May 2020

The ICPC is an annual multi-tiered competitive programming competition among universities around the world. As Chief Technical Staff, I led the team to:

- **build contest-customized OS images** and distribute them on contestant hosts using **Network boot**.
- **restrict external access and host-to-host access** of each contestant's host device.

PROJECTS

shmem-bind

Author and maintainer of shmem-bind, an open-source Rust library that provides a safe, idiomatic wrapper over shared memory APIs with proper cleanup. (git)

ARK

A Layer-4 VPN protocol that hides traffic from an adversary by switching between Layer-4 protocols from hub to hub and making inbound and outbound traffic unbalanced. (git)

Cell Network Simulator

A 4G/5G Cell Network Simulator that virtualizes the entire Layer 1–3 stack. The system provides seamless, transparent integration, allowing users to route arbitrary TCP/UDP traffic through the simulation via standard commands (e.g., curl) without any modifications to the Layer 4 application or benchmarks. (git)

CS162 Berkeley's Pintos Project

Developed an operating system kernel, implementing system calls, file system management and process scheduling. (git)

Sparse Matrix-Vector Multiplication with AVX

Developed a sparse matrix-vector multiplication library that uses Intel AVX vector instructions for higher performance. (git)

TCP Network Stack

Implemented the TCP state machine from RFC 793 in Rust, on top of the Linux kernel's TUN/TAP interface. (git)

C-minus Compiler

Implemented a compiler for the C-minus language. (git)

Hash Map Based on Split-Ordered Lists

Implemented a concurrent hash map in Rust based on the split-ordered lists paper. (git) (paper)

Hazard Pointers

Implemented hazard pointers in Rust for safe memory reclamation. (git)

SELECTED COURSES

- Operating Systems 20/20
- Computer Networks 20/20
- Database Design 20/20
- Compiler Design 20/20
- Automata Theory 20/20
- Computer Architecture 20/20
- Multi-Core Computing 20/20
- Real-time Systems 20/20

INDUSTRIAL EXPERIENCE (SUMMARIZED)

Software Engineer

Sotoon

Jan 2023 – Dec 2024

Sotoon is a B2B cloud provider that focuses on reliable and scalable computing solutions. I worked in the K8s-as-a-Service team, designing and maintaining Sotoon's Kubernetes engine and several integrations to Sotoon's services, such as CSI, CNI, and Auto-scaler.

Technology Stack: Golang, Kubernetes, CSI, CNI, Docker, Calico, Ansible, Packer

Data Platform Engineer

Digikala

Aug 2021 – Jan 2023

Digikala is the largest e-commerce company in the Middle East, serving over 40 million users. As part of the Big Data & AI team, I was responsible for building and maintaining the infrastructure required to efficiently store, process, and manage the vast amounts of data generated by user activity.

Technology Stack: Rust, Spark, Airflow, Kubernetes, Ceph, Kafka, Debezium, GitOps, ArgoCD

DevOps Engineer

Tapsell

Nov 2019 – Aug 2021

Tapsell is a leading AdTech company in the Middle East. As part of the DevOps team, I provided cloud infrastructure and platform services that enabled other technical teams to build, deploy, and scale their applications efficiently.

Technology Stack: Kubernetes, Linux, Docker, Cassandra, MongoDB, Kafka, Redis

PROGRAMMING LANGUAGES

Rust | C | Go | OCaml | Python | Java | SQL | Bash

TECHNOLOGIES

Parallel Computing

AVX | CUDA | OpenMP | rayon-rs | crossbeam-rs

Orchestration

k8s | Docker | Docker-swarm

Networking

Nginx | HAProxy | Calico

CI/CD

Ansible | Terraform | GitLab CI | ArgoCD

Bigdata

Spark | Airflow

Database & Storage Systems

Ceph | MinIO | Kafka | Redis | Cassandra | MongoDB

LANGUAGES

English ●●●●●

French ●●●○○

Persian ●●●●●